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A hybrid model for detecting motion artifacts in ballistocardiogram signals

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Abstract

Background: The field of contactless health monitoring has witnessed significant advancements with the advent of piezoelectric sensing technology, which enables the monitoring of vital signs such as heart rate and respiration without requiring direct contact with the subject. This is especially advantageous for home sleep monitoring, where traditional wearable devices may be intrusive. However, the acquisition of piezoelectric signals is often impeded by motion artifacts, which are distortions caused by the subject of movements and can obscure the underlying physiological signals. These artifacts can significantly impair the reliability of signal analysis, necessitating effective identification and mitigation strategies. Various methods, including filtering techniques and machine learning approaches, have been employed to address this issue, but the challenge persists due to the complexity and variability of motion artifacts.

Methods: This study introduces a hybrid model for detecting motion artifacts in ballistocardiogram (BCG) signals, utilizing a dual-channel approach. The first channel uses a deep learning model, specifically a temporal Bidirectional Gated Recurrent Unit combined with a Fully Convolutional Network (BiGRU–FCN), to identify motion artifacts. The second channel employs multi-scale standard deviation empirical thresholds to detect motion. The model was designed to address the randomness and complexity of motion artifacts by integrating deep learning capabilities with manual feature judgment. The data used for this study were collected from patients with sleep apnea using piezoelectric sensors, and the model's performance was evaluated using a set of predefined metrics.

Results: This paper proposes and confirms through analysis that the proposed hybrid model exhibits exceptional accuracy in detecting motion artifacts in ballistocardiogram (BCG) signals. Employing a dual-channel approach, the model integrates multi-scale feature judgment with a BiGRU–FCN deep learning model. It achieved a classification accuracy of 98.61% and incurred only a 4.61% loss of valid signals in non-motion intervals. When tested on data from ten patients with sleep apnea, the model demonstrated robust performance, highlighting its potential for practical use in home sleep monitoring.

Conclusion: The proposed hybrid model presents a significant advancement in the detection of motion artifacts in BCG signals. Compared to existing methods such as the Alivar method [29], Enayati method [22], and Wiard method [20], our hybrid



model achieves higher classification accuracy (98.61%) and lower valid signal loss ratio (4.61%). This demonstrates the effectiveness of integrating multi-scale standard deviation empirical thresholds with a deep learning model in enhancing the accuracy and robustness of motion artifact detection. This approach is particularly effective for home sleep monitoring, where motion artifacts can significantly impact the reliability of health monitoring data. The study findings suggest that the proposed hybrid model could serve as a valuable tool for improving the accuracy of motion artifact detection in various health monitoring applications.

Keywords: Piezoelectric sensing, Motion artifact detection, Hybrid model, Sleep monitoring

Background

Owing to the progress in sensing technology, piezoelectric sensing has emerged as a potent instrument for contactless monitoring of vital signs [1–3], capable of efficiently capturing critical life sign signals like heart rate and respiration while the subject is at rest [4]. Nevertheless, when monitoring sleep vital signs with piezoelectric sensors, the process is frequently disrupted by motion artifacts. Motion artifacts arise as signal distortions in the collection and processing of biomedical signals, influenced by external environmental factors and the subject's bodily movements. Such interferences can blend with physiological features within the biological signals, making the accurate interpretation of the signals more complex [5–7]. Motion artifacts are prevalent across various types of vital sign monitoring devices, significantly impairing the reliability of signal analysis. The motion artifacts during sleep originate from involuntary muscle tremors and body movements such as turning over at night. As an example of wearable heart rate monitoring, numerous approaches to artifact detection and suppression have been reported, including linear filtering [1], adaptive filtering [2], wavelet denoising [3], Bayesian filtering techniques [4], statistical approaches [8, 9], independent component analysis [10], empirical mode decomposition [10–12], and Blind source separation [13], among others. In addition to the aforementioned methods, a variety of machine learning and deep learning techniques have been proposed to address the issue of motion artifacts in ballistocardiogram (BCG) signals. A Recurrent Neural Network (RNN) based on Long Short-Term Memory (LSTM) was proposed for denoising electrocardiogram (ECG) signals [14]. In addition, a fully connected denoising autoencoder is proposed for ECG signal denoising [15]. Pradhan proposed a technique based on Generative Adversarial Networks (GANs) to suppress artifacts in ECG signals [16, 17]. Furthermore, a Deep Recurrent Neural Network (DRNN) based on dual-LSTM was proposed, which incorporates an attention mechanism for suppressing motion artifacts and baseline wandering [18]. In another study, a deep neural network model based on the sparse representation of signals is presented for ECG signal denoising [19]. Krishnan et al. [20] proposed a two-stage photoplethysmogram (PPG) data processing method based on higher order statistical parameters, with the detection unit employing the Neyman–Pearson detection criterion and motion suppression using a combination of frequency-domain independent component analysis (FD-ICA) and periodicity estimation. Furthermore, the removal of motion artifacts from ECG and PPG signals has garnered increased attention [21, 23–27]. The method leverages the characteristics of two PPG signals: the high correlation of the motion artifacts (MA) and their distinct AC/

DC ratios [28]. These characteristics allow the adaptive filter to cancel the MA while preserving the AC component. In addition, the adaptive filter employs the sign-sign least mean squares (SS-LMS) algorithm to minimize hardware resource requirements. Compared to wearable devices, piezoelectric-based sleep monitoring devices are more susceptible to motion artifacts. To address this issue, Enayati et al. [22] employed a machine learning-based multi-feature extraction method, extracting 53 features from both time and frequency domains, including standard deviation (STD), median absolute deviation (MAD), 75th percentile, skewness, kurtosis, and Shannon entropy, and utilized support vector machines (SVM) and the RUSBoost algorithm for the classification of motion artifacts.

Although multi-feature extraction methods hold promise for enhancing the detection of motion artifacts, they often introduce increased complexity and challenges in generalizing across multi-dimensional feature spaces, necessitating further validation. To address these issues, Alivar et al. [29] proposed a simplified motion artifact detection rule based on the mean and variance of fixed-scale data. This method uses an expected probability threshold to classify each 1-s segment of the collected vital signs as either normal signals or motion artifacts, effectively leveraging the variance (or standard deviation, STD) to detect motion artifacts. Several methods have been proposed for motion artifact detection in BCG signals. For example, Alivar et al. [29] introduced a sequential detection method using upper and lower thresholds to identify motion artifacts. Enayati et al. [22] employed a machine learning approach with support vector machines (SVM) and feature extraction for classification. Wiard et al. [20] developed an automatic detection method using frequency-domain independent component analysis (FD-ICA) and periodicity estimation for BCG signals measured on a modified bathroom scale. While these methods have shown promise, they often face challenges in terms of accuracy and computational efficiency. There remains a need for a more accurate and efficient approach to motion artifact detection in BCG signals.

While the aforementioned methods are effective for most motion artifact detection, due to the complexity and randomness of motion artifacts (manifested in aspects such as signal amplitude, envelope shape, duration, and distribution), taking as an example the cardiac and respiratory vibration signals obtained by piezoelectric sensing, as shown in Fig. 1a and b, traditional methods for motion artifact detection based on empirical thresholds or features still pose significant challenges in actual motion artifact detection.

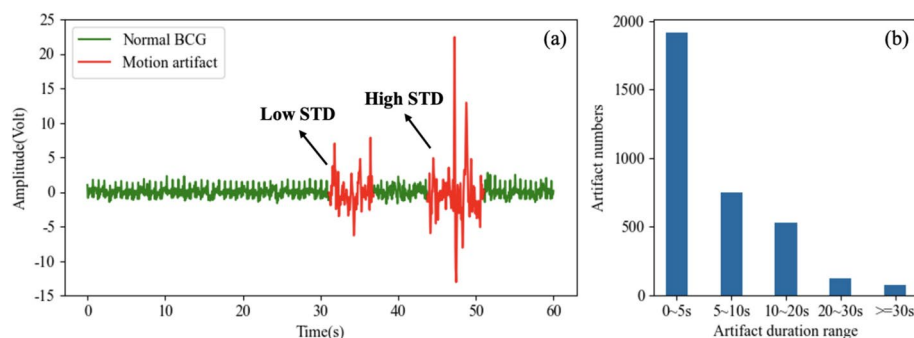


Fig. 1 **a** Signal segments containing motion artifacts of varying severity. **b** Distribution of the duration of motion artifacts in the dataset used for performance validation in this study

To address the issue of detection accuracy being insufficient due to the randomness of motion artifacts, this paper proposes a novel hybrid motion artifact detection scheme that integrates multi-scale decision rules with deep learning. Specifically, multi-scale STD is used as the motion judgment criterion, addressing the limitation of single fixed-scale features in failing to comprehensively cover various types of artifact segments. Meanwhile, the deep learning model enhances the abstraction of feature dimensions, effectively extracting abstract features under significant differences in the morphology and duration of motion artifact interference. This overcomes the limitations of traditional manually crafted features with limited dimensions in detecting motion artifacts, thereby improving the accuracy and robustness of detection.

To validate the accuracy and effectiveness of the hybrid model in motion artifact detection, this study analyzed nocturnal vital sign data obtained via piezoelectric sensing from ten patients with Sleep Apnea Syndrome. The experimental results show that using the method proposed in this study, the classification accuracy for motion artifacts reached 98.61%, with a 4.61% effective signal loss in non-motion intervals, validating the superior performance of this method in motion artifact detection.

Results

A. Experimental setup

The deep neural network model in this study was constructed using the PyTorch framework in Python. The workstation used for this study is equipped with an Intel Core Ultra7 165H processor with a maximum turbo frequency of 5.0 GHz and 32 GB of RAM. The deep learning computations are accelerated by an NVIDIA GeForce RTX A2000 GPU.

B. Evaluation metrics

The performance evaluation of the detection effectiveness is based on the ideas of object detection evaluation [30, 31]. In this study, the detection rate is adopted as the metric to measure the number of body movements detected. In addition, drawing from the concept of Intersection over Union (IoU) [30, 32], the ratio of signals that are falsely detected (outside the intersection of true and detected labels for body movement artifacts) to the effective signal loss is used as the false detection rate metric for evaluating the detection algorithm.

The detection rate is defined as:

$$R_{chk} = \frac{N_{chk}}{N_{all}} \quad (1)$$

The detection rate R_{chk} is defined as the number of detected body motion artifacts divided by the total number of motion artifacts, where N_{chk} is the number of true labels detected and N_{all} is the total number of true labels.

The detection is defined as: the start and end range of a true motion artifact label being completely contained by a detected motion artifact label, or the part not contained is relatively small (the average duration of the start and end positions not being contained is less than 2 s, provided that the central position of the true label is included by the detected label).

The ratio of valid signal loss is defined as:

$$L_{effect} = \frac{t_{chk} - t_{chk_lb}}{t_{all} - t_{lb}} \quad (2)$$

The valid signal loss ratio L_{effect} is defined as the total duration of all false positive segments excluding the detected true labels, as a proportion of the total duration of valid signals in the subject of entire night signals (i.e., the total duration of the night signals minus the total duration of body motion artifacts). Here, t_{chk} represents the total duration of all detected labels, and t_{chk_lb} is the total duration of the true labels detected. Subtracting the latter from the former yields the combined duration of the over-detected segments around the true labels (at the start and end) as well as the sum of durations of the remaining over-detected labels.

t_{all} is the total duration of all signals for the subject of entire night, t_{lb} is the total duration of all true labels, subtracting the latter from the former gives the total duration of valid signals for the entire night.

In addition, the classification accuracy of the deep learning model, denoted as accuracy, is defined as:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

The recall rate of the model, denoted as recall, is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

The precision rate of the model, denoted as precision, is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

The F1 score is defined as:

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (6)$$

C. Verification of BiGRU-FCN network architecture and performance comparison of different models

In time series classification research, common network models include Temporal Convolutional Networks (TCN) [33], Fully Convolutional Networks (FCN) [34], Residual Neural Networks (ResNet) [35], Long Short-Term Memory (LSTM) networks [36], Gated Recurrent Units (GRU) [37], InceptionTime [38], and LSTM Fully Convolutional Network[44], among others. These models were applied to the body movement artifact detection classification test in this study, and their performances are presented below.

As shown in Table 1, the BiGRU-FCN model proposed in this study exhibits the best classification performance and has a moderate number of parameters. Regarding the model of classification speed, we conducted experiments using the ten overnight test samples detailed in Table 5 as our experimental dataset. Without relying on GPU

Table 1 Performance comparison and verification of different network architectures

Model	Accuracy	Recall	Precision	F1 score	Parameter
TCN[33]	0.9643	0.9579	0.9687	0.9633	166,502
FCN[34]	0.9884	0.9865	0.9896	0.9881	711,426
ResNet[35]	0.985	0.983	0.9862	0.9846	733,954
LSTM[36]	0.9667	0.9651	0.9664	0.9658	241,002
GRU[37]	0.9612	0.9621	0.9584	0.9603	180,802
InceptionTime[38]	0.9894	0.9881	0.9901	0.9891	550,914
LSTM-FCN[44]	0.9887	0.9894	0.9875	0.9884	1,193,426
Our proposed	0.9931	0.993	0.9928	0.9929	720,066

Table 2 Verification of detection performance using statistical feature threshold methods at various scales

Scale	Signal loss statistics	Detection rate statistics
0.5 s	24.23%	89.74%
1 s	11.14%	95.09%
2 s	10.27%	92.08%
3 s	10.56%	87.65%
4 s	11.26%	83.78%
5 s	12.48%	81.79%

acceleration, the model achieved an average classification time of approximately 26 ms for each 5-s segmented signal. When processing an entire hour of data, the model leverages batch processing and other optimizations, resulting in an average processing time of only 94 μ s per segment. This demonstrates the high efficiency and timeliness of the model when handling large volumes of data.

D. Verification of the performance of the statistical feature threshold decision channel

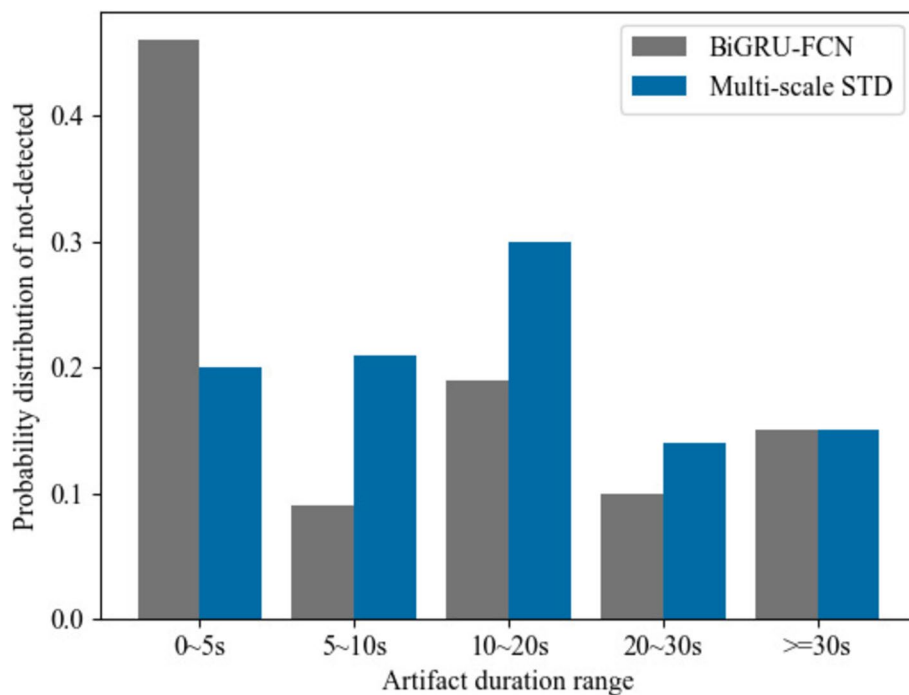
For the proposed decision channel based on global and local standard deviations for body movement artifact detection, we examine its performance at different scales and validate the parameter configuration that yields the highest detection rate. As shown in Table 2, the detection rate of this channel is highest at a scale of 1 s; therefore, a scale of 1 s is selected for subsequent tests.

E. Verification of the performance of the dual-channel fusion architecture

Based on the aforementioned BiGRU–FCN network model and dual-channel statistical features, this section validates the performance of the entire architecture that integrates the decision layers of both and incorporates post-processing algorithms. As shown in Table 3, the overall performance of the framework, in terms of both detection rate and the loss of valid signals, is superior to that of a single-channel method, thereby confirming the effectiveness of the method proposed in this study.

Table 3 Overall performance verification of the algorithm under dual-channel fusion and post-processing

Serial number	Motion artifact count	BiGRU-FCN L_{effect}	BiGRU-FCN R_{chk}	Handcrafted features L_{effect}	Handcrafted features R_{chk}	Post-fusion L_{effect}	Post-fusion R_{chk}	Post-fusion Recall
1	335	7.72%	90.55%	7.88%	92.99%	6.08%	99.39%	97.57%
2	216	5.94%	91.36%	3.22%	97.27%	1.81%	99.09%	93.66%
3	223	9.16%	90.99%	13.35%	93.56%	7.76%	99.57%	97.27%
4	87	2.03%	89.66%	1.08%	96.55%	0.75%	97.70%	93.66%
5	124	4.32%	86.99%	11.46%	93.50%	3.63%	97.56%	95.32%
6	214	12.91%	92.06%	17.49%	91.59%	6.72%	97.66%	87.82%
7	275	8.84%	91.55%	21.25%	98.59%	6.82%	100.00%	95.12%
8	742	18.52%	95.01%	13.54%	97.71%	4.60%	98.11%	82.04%
9	515	16.51%	90.10%	15.58%	95.34%	4.85%	99.03%	95.50%
10	353	9.76%	90.37%	6.56%	93.77%	3.12%	98.02%	84.91%
Mean	308.4	9.57%	90.86%	11.14%	95.09%	4.61%	98.61%	92.29%
SD	185	4.91%	1.92%	6.07%	2.22%	2.18%	0.85%	5.13%

**Fig. 2** Comparison of the probability distribution of undetected body movement artifacts by each channel along the duration dimension

F. Analysis of the complementarity of the dual channels

As shown in Table 6, the performance of the hybrid model is superior to that of a single detection method, especially in terms of detection rate, due to the complementary effect between the two channels after fusion. To explore the strengths and weaknesses of each dual channel, we conducted a statistical analysis of the duration of motion artifacts that were not detected. As shown in Fig. 2, in the interval of 0 to 5 s, the false negative rate of

the deep learning model is significantly higher than that of the manual feature method. In the detection of segments longer than 5 s, the deep learning model performs better overall than the manual feature method. This is because the longer the duration of body movement, the more diverse the features become (with stronger randomness in signal features), making it more difficult for fixed manual features (even multi-scale) to effectively detect body movements. Therefore, deep learning models have an advantage due to their stronger capability to extract abstract features from complex signals. In contrast, shorter duration body movement artifacts have relatively simpler features, so the multi-scale manual feature method performs better than the deep learning model.

G. Comparative analysis with existing motion artifact detection methods

To further validate the effectiveness of the proposed hybrid model, we compared its performance with several existing methods specifically designed for motion artifact detection in BCG signals. The methods selected for comparison include the Wiard Method [20], the Enayati Method [22], and the Alivar Method [29]. These methods have been widely recognized for their contributions to motion artifact detection and provide a robust benchmark for evaluating the proposed approach. The performance metrics used for comparison are classification accuracy, detection rate, false detection rate, and valid signal loss ratio. The results are summarized in Table 4.

Conclusion

The proposed hybrid model presents a significant advancement in the detection of motion artifacts in BCG signals. Compared to existing methods such as the Alivar et al. [29], Enayati et al. [22], and Wiard et al. [20], our hybrid model achieves higher classification accuracy (98.61%) and lower valid signal loss ratio (4.61%). This demonstrates the effectiveness of integrating multi-scale standard deviation empirical thresholds with a deep learning model in enhancing the accuracy and robustness of motion artifact detection. This approach is particularly effective for home sleep monitoring, where motion artifacts can significantly impact the reliability of health monitoring data. The study findings suggest that the proposed hybrid model could serve as a valuable tool for improving the accuracy of motion artifact detection in various health monitoring applications.

Methodology

A. Data acquisition and preprocessing

The piezoelectric sensing health monitoring device deployed in this study is provided by Guangzhou Zhongke New Knowledge Technology Co., Ltd., consisting of piezoelectric

Table 4 Comparative analysis of motion artifact detection methods

Model	Accuracy	Detection rate	False detection rate	Valid signal loss ratio
Wiard et al. [20]	96.00%	94.00%	6.00%	8.00%
Enayati et al. [22]	94.50%	93.00%	8.00%	10.00%
Alivar et al. [29]	92.00%	90.00%	10.00%	12.00%
Our Proposed	98.61%	95.09%	4.61%	4.61%

sensor modules and data processors with a sampling frequency of 1 kHz. During measurement, the piezoelectric sensor module is placed under the pillow. As the human body is a micro-vibration entity, piezoelectric sensors can capture the micro-vibration forces caused by the heart's contraction and relaxation, as well as the chest expansion movements during the breathing process, namely the cardio ballistocardiogram (BCG) signals and respiratory effort. The mixed analog signals are amplified, noise is removed, and then converted into digital signals through a 12-bit analog-to-digital converter (ADC). ECG and BCG signals are recorded simultaneously at the same sampling frequency using BIOPAC MP160. Before data analysis, both ECG and BCG channel signals have been synchronized, with ECG serving as a reference for BCG signal analysis.

In this study, the signal structural preprocessing flow mainly includes: signal downsampling, industrial frequency and high-frequency noise filtering, windowing segmentation, Z score normalization, and global mean removal. The main process flowchart of structural preprocessing is shown in Fig. 3

(1) Signal downsampling

The clinical subject's entire night vital signs signal (hereinafter referred to as the raw signal), acquired by the non-intrusive sensing device with a sampling rate of 1 kHz, is subjected to signal downsampling. A method of uniform interval sampling is used to obtain a signal with a lower sampling rate. For motion artifact detection, the main observational features include the basic waveform shape, the temporal sequence relationship before and after, the amplitude, etc. A higher sampling rate does not affect these indicators, but rather reduces the speed of signal computation and processing, affecting the efficiency of program

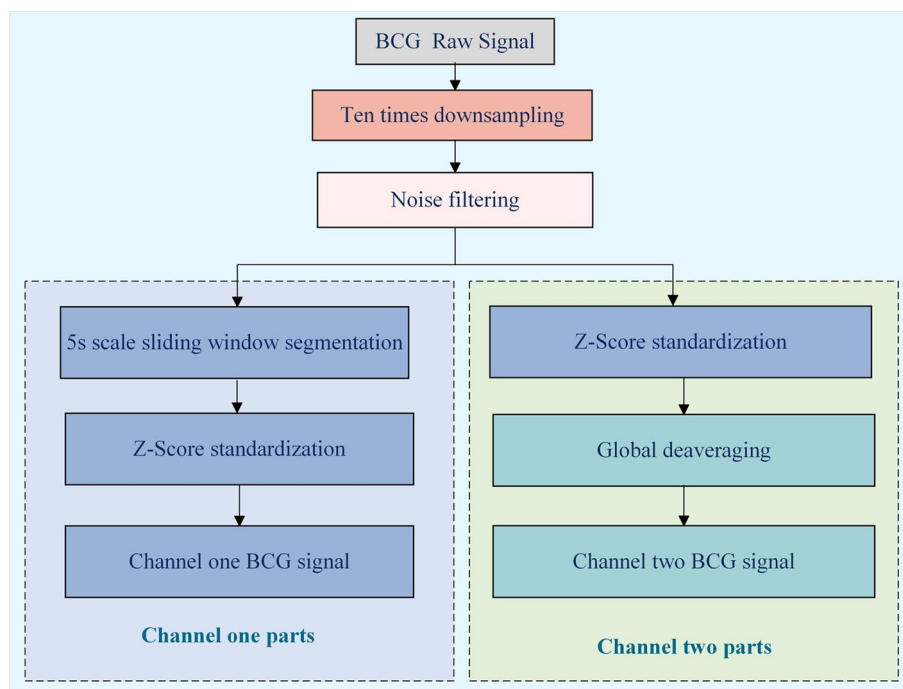


Fig. 3 Signal structural preprocessing flowchart

execution. Therefore, the sampling rate is reduced by a factor of ten, resulting in a signal frequency of 100 Hz after downsampling.

(2) Noise filtering

The raw signal collected by the non-intrusive sensing device is a superimposed signal, including BCG signals, respiratory signals, power frequency noise, high-stress random noise, etc. Since the main frequency distribution of the BCG signal is below 10 Hz, to remove high-frequency noise, this plan designs a 6th order Butterworth low-pass filter with a cutoff frequency of 10 Hz, used to remove high-frequency noise, retaining only the BCG signal and respiratory signal components.

(3) Sample window segmentation

For single-scale deep learning models, the length of each data input is fixed. Deep learning algorithms can simulate human learning of “empirical knowledge” to some extent, that is, simulate human observation and recognition of motion artifact signals at a certain timescale. The normal heart rate for humans is commonly between 30 and 120 beats per minute (i.e., 0.5-2 Hz), and to distinguish normal signals from motion artifacts, the human eye observation timescale should be greater than one heartbeat timescale (0.5 Hz i.e., 2 s for one heartbeat), otherwise a complete BCG signal cannot be observed, hence the model learning scale must be greater than or equal to 2 s. For human eye recognition, the larger the scale, the more effective information can be observed at the same time. However, for machine recognition, the larger the single recognition scale, the easier it is to waste more effective signals (because quite a few motion artifacts only last for a few seconds). Therefore, based on test experience, 5 s is chosen as the model training and testing scale.

(4) Z score normalization

For deep learning models, the task is to learn the sequential and morphological differences between normal signals and motion artifact segments. Since BCG signals have individual differences, different subjects' collected data will have differences in the range of amplitude fluctuations, whether it is normal signals or motion artifact segments. Therefore, in the signal preprocessing process, it is necessary to reduce the differences in the amplitude fluctuation range between different subjects as much as possible, ensuring that the amplitude fluctuation range of normal signals and motion artifact segments between different subjects is close, and not affecting the original temporal sequence relationship and waveform shape of the signals. Z score normalization subtracts the mean value of the signal and then divides by the standard deviation, making the signal fluctuate around 0, and the fluctuation range is more unified (the standard deviation equals 1), so this method can meet the requirements very well.

$$Z = \frac{x - \mu}{\sigma} \quad (7)$$

where x is the input signal, μ is the mean value, and σ is the standard deviation.

(5) Global mean removal

Unlike the deep learning model of channel one, channel two needs to learn the temporal relationship and morphological relationship of normal signals and motion artifact signals through stable amplitude range data, while channel two directly extracts global and local multi-scale standard differences, and then sets empirical threshold values for motion artifact discrimination, so it is necessary to retain the standard deviation characteristics of the original signal. Therefore, only the mean removal operation in Z score is performed for channel two, as shown in Fig. 3, making the original signal fluctuate around 0 after mean removal, reducing the impact of the overall signal baseline on the discrimination.

B. Experimental data and motion artifact annotation rules

(1) Experimental data

The experimental data in this study were provided by the National Key Laboratory of Respiratory Disease in Guangzhou. The study was conducted under the approval of the Academic Committee of South China Normal University (Approval No.: SCNU-PHY-2021-053). Utilizing piezoelectric sensing, heart and respiratory data from 59 patients with sleep apnea were collected from 10:00 PM to 7:00 AM the following day. All subjects have provided their informed consent. Concurrently with the data collection, subjects collected three-lead ECG to serve as a reference for assessing the efficacy of the piezoelectric signals. There were 52 males and 7 females, aged between 10 and 75 years old. There are 9 healthy subjects, and the patients with mild, moderate, and severe sleep apnea are 18, 9, and 23, respectively, with Apnea–Hypopnea Index (AHI) values ranging from 0.5 to 97.4 events per hour.

2. Motion artifact annotation rules

Consistent with references [22, 39, 40], the presence of motion artifacts in the sensing data is determined using a 5-s sliding time window for data segmentation, with a 1-s sliding step as shown in Table 5. When a signal segment of amplitude range exceeds twice the amplitude range of the surrounding normal segments, and is deemed unrecognizable by manual adjudication within that scale, the signal at that scale is defined as a motion artifact. Figure 4 shows an example of motion artifact annotation at a certain time scale, where the red signal indicates the motion-annotated area, and the green signal represents the non-motion area (normal vital sign signal).

Table 5 Sample distribution of normal signal and motion artifact datasets

Category	Description	Number of segments
Positive samples (normal signal)	Segments classified as normal signals	120,924
Negative samples (motion artifact)	Segments classified as motion artifacts	115,373
Total number of segments	Total number of 5-s segments in the dataset	236,297

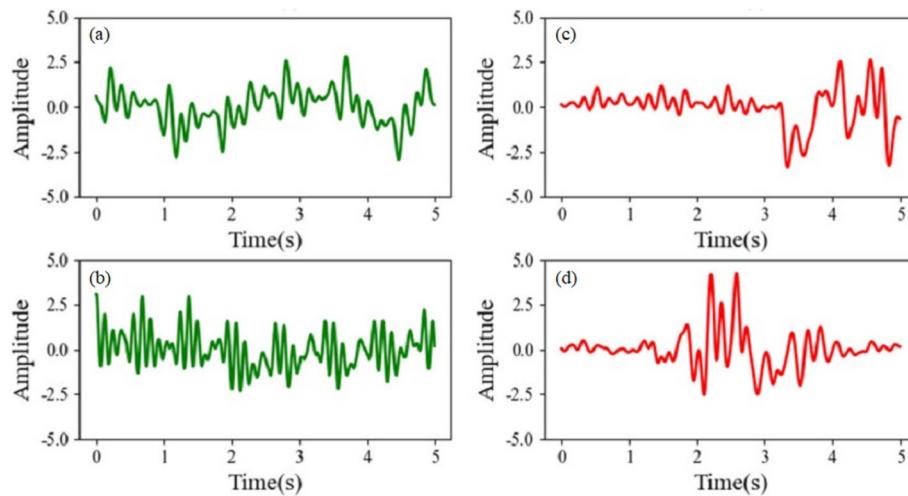


Fig. 4 Sample segmentation results at a 5-s scale, where (a) and (b) are positive samples (normal signals), and (c) and (d) are negative samples (motion artifacts)

Considering the randomness of motion artifacts, manual annotation of motion signals should begin with non-motion areas. According to numerical statistics of the dataset for patients with Sleep Apnea Syndrome (SAS), the duration of motion artifacts is primarily within 10 s. Therefore, this study adopts a 5-s sliding window for data segmentation and annotation. The specific process involves starting from the beginning of the motion, using a 5-s scale sliding window, advancing 1 s until the end of the motion, within the aforementioned time scale, each 5-s segment is annotated as motion (labeled as “1”). Similarly, non-motion area signals are annotated as “0” following the same process.

C. Motion artifact detection

(1) Hybrid model architecture

The hybrid model architecture for motion artifact detection proposed in this paper is shown in Fig. 5. Addressing the randomness of motion artifact signals in the two-dimensional space of duration and amplitude, a combined detection approach utilizing both handcrafted features and deep network judgments is employed. Based on multi-scale STD decision-making, the model integrates the deep learning model of ability to express abstract features of random motion signals, enhancing the accuracy of motion detection.

2. Multi-scale STD motion artifact detection

Under normal circumstances, the body movement artifact exhibits a significantly increased level of randomness and disorder in its envelope compared to the normal signal. Therefore, the global and local relative standard deviations of the signal at different time scales are utilized as an empirical basis for determining significant fluctuations, complementing the BiGRU-FCN body motion detection.

More specifically, this paper introduces an empirical model for body movement detection within the hybrid model, utilizing a joint decision approach based on

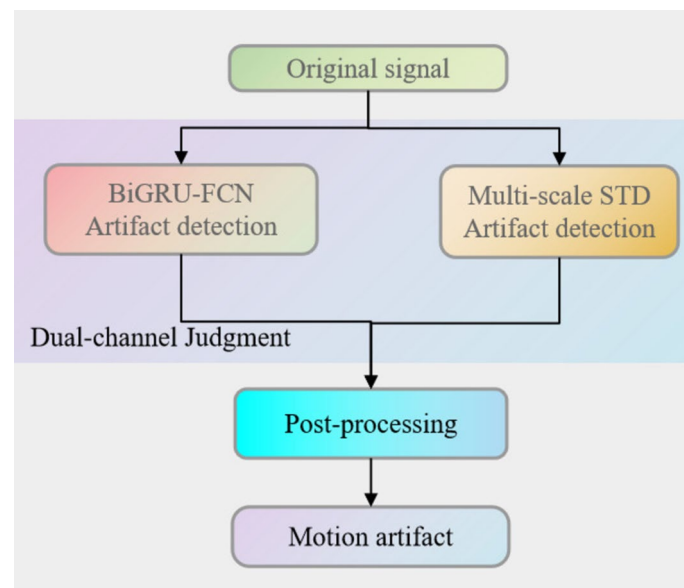


Fig. 5 Overall flowchart of this algorithm

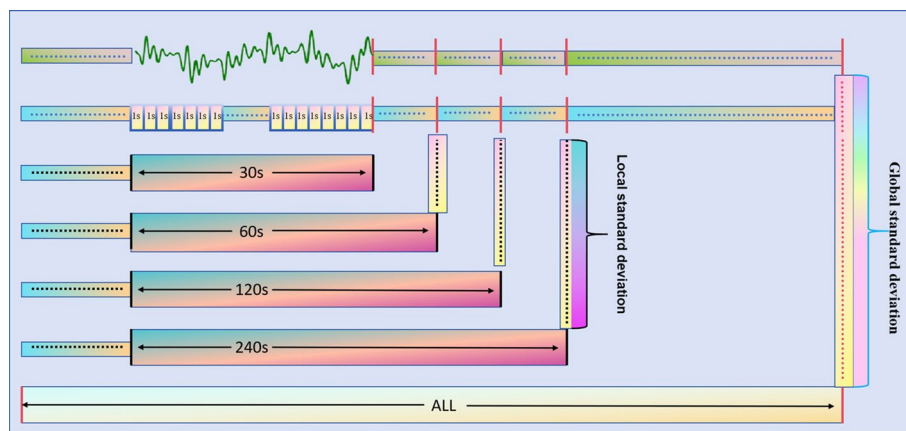


Fig. 6 The schematic diagram for extracting global and local standard deviations from Channel 2

multi-time scale standard deviations. The global standard deviation is defined as the median of all nighttime signals from the subject, calculated using 1-s time scale standard deviations, as depicted in Fig. 6. Meanwhile, the local standard deviation represents the median of standard deviations for each 1-s signal at various time scales ($T = 30, 60, 120, 240$ s), as detailed in Table 6. If the minimum value among these five medians exceeds twice the current 1-s signal of standard deviation, then that particular second is classified as a body movement artifact.

(3) Motion artifact detection based on BiGRU-FCN

Due to the stochastic nature of motion artifact interference, the proposed hybrid detection model in this study employs a deep learning architecture that integrates

Table 6 Characteristics of multi-scale artificial experience

Features	Description
T1	The median of 1-s standard deviations throughout the entire night
T2	Compute the median of all 1-s standard deviations within the 30-s window following the current 1-s segment
T3	Compute the median of all 1-s standard deviations within the 60-s window following the current 1-s segment
T4	Compute the median of all 1-s standard deviations within the 120-s window following the current 1-s segment
T5	Compute the median of all 1-s standard deviations within the 240-s window following the current 1-s segment

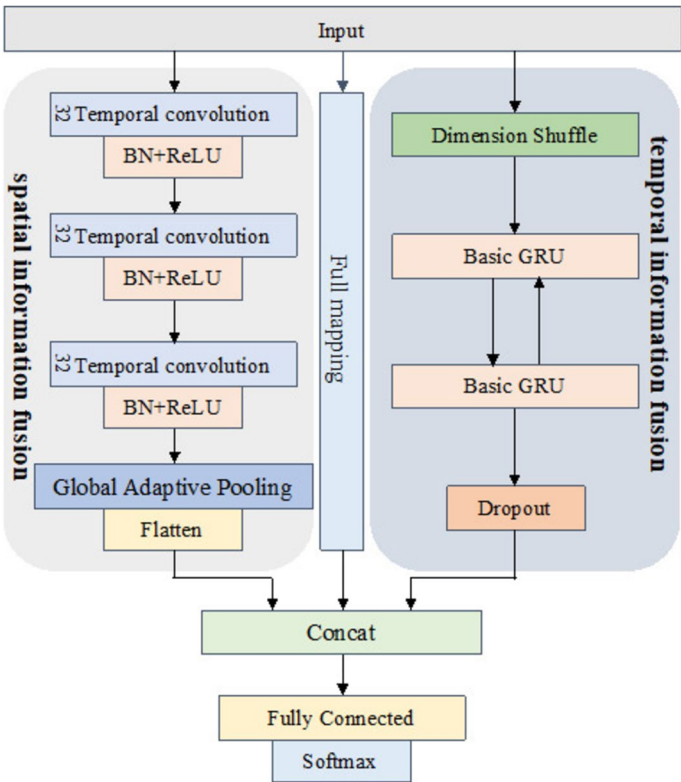


Fig. 7 Flow chart of the BiGRU-FCN network architecture

RNN and CNN [26, 27], known as BiGRU-FCN. As shown in Fig. 7, the BiGRU-FCN designed in this paper includes three channels of input, which are used to learn the time sequence features, high-dimensional abstract features, and low-dimensional morphological features of the movement with the body, respectively.

Specifically, the model of RNN architecture employs BiGRU, Bidirectional Gated Recurrent Units [41], to capture the sequential features of vital sign parameters in both forward and backward directions. Compared with the LSTM (Long Short-Term Memory) network, BiGRU features fewer parameters [26, 42]. To learn the high-dimensional abstract features of motion artifacts, the model of CNN architecture is composed of three convolutional layers [7]. In this model, the sizes of the convolutional kernels are set

sequentially from largest to smallest as $7*1$, $5*1$, and $3*1$, to effectively extract abstract semantic information from the signal across different scales.

To enhance the model of capability to learn superficial morphological features of vital sign signals, the model input incorporates a full mapping of an MLP (Multilayer Perceptron) [43], utilized for directly mapping input signals to classification outputs.

(4) Fusion and post-processing of motion artifact detection

It can be inferred that, based on the BiGRU–FCN and multi-scale standard deviation, the classification output time scale for motion artifacts is 1 s, the predicted output of the BiGRU–FCN model at the t -th second is defined as $f(t)$, and the detection decision result of the multi-scale standard deviation at the t -th second is $g(t)$, for $t = 1, 2, \dots, T$ (seconds), the detection result of the hybrid model is $y(t) = f(t) \cup g(t)$, where the symbol $\cup$ represents the OR operation.

Nevertheless, simply integrating the BiGRU–FCN with the multi-scale standard deviation, decision results may to a certain extent lead to false positives in body motion artifact detection, where normal signals are mistakenly judged as body motion. Therefore, this study proposes a post-processing scheme based on empirical rule constraints to perform a secondary screening of the integrated motion interference, thereby reducing the probability of false positives in body motion artifact detection.

The detailed screening criteria are as follows:

For each 1-s segment within the motion artifact label, obtain the standard deviation std_i and the secondary standard deviation std^2 (re-calculate the standard deviation based on the values of each standard deviation to assess signal variability), compare with the predetermined empirical thresholds; segments exceeding the threshold limits are considered motion artifacts, while others are deemed normal signals.

$$y_i = \begin{cases} 1 & \text{if } std_i < \theta_1 \text{ and } std^2 < \theta_2 \\ 0 & \text{if } std_i > \theta_1 \text{ and } std^2 < \theta_2 \\ 0 & \text{if } std^2 > \theta_2 \end{cases} \quad (8)$$

where θ_1 is calculated from the standard deviation of the normal signals before and after the motion artifacts, and θ_2 is set based on empirical thresholds.

Abbreviations

BCG	Ballistocardiogram
ECG	Electrocardiogram
BiGRU–FCN	Bidirectional Gated Recurrent Unit combined with a Fully Convolutional Network
RNN	Recurrent neural network
LSTM	Long Short-Term Memory
GANs	Generative Adversarial Networks
DRNN	Deep Recurrent Neural Network
FD-ICA	Frequency-domain independent component analysis
PPG	Photoplethysmogram
MA	Motion artifacts
SS-LMS	Sign-sign least mean squares
STD	Standard deviation
MAD	Median absolute deviation
SVM	Support vector machines
AHI	Apnea–Hypopnea Index
SAS	Sleep Apnea Syndrome
CNN	Convolutional neural network
MLP	Multilayer perceptron

IoU	Intersection over Union
TP	True positive
FP	False positive
TN	True negative
FN	False negative
TCN	Temporal Convolutional Networks
FCN	Fully Convolutional Networks
ResNet	Residual Neural Networks
GRU	Gated Recurrent Units

Author contributions

YLJ developed the idea for this study. HZ and QZ discussed and performed the statistical analysis to prove the availability of this study. All authors read and approved the final manuscript.

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Data availability

No datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate

The study was conducted under the approval of the Academic Committee of South China Normal University (Approval No.: SCNU-PHY-2021-053). All subjects involved in this study have provided their informed consent.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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